

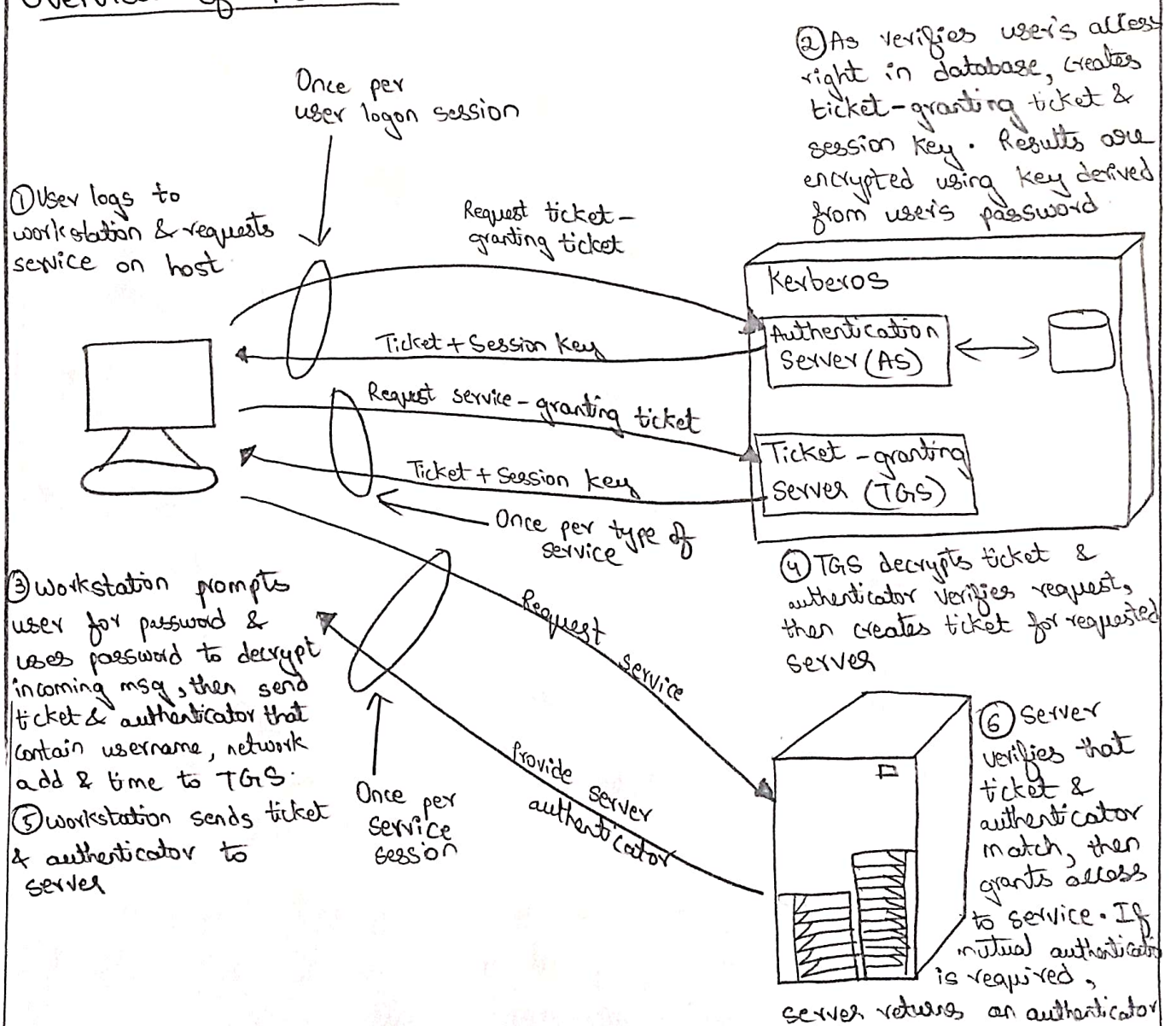
ASSIGNMENT - 03

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1. Explain the overview of Kerberos differentiating Kerberos Version 4 and version 5?

Ans: Kerberos is a network authentication protocol designed to provide secure authentication for users and services within a distributed system, primarily over an insecure network. It uses symmetric-key cryptography & a trusted 3rd party known as the Key Distribution Center (KDC), to authenticate entities.

Overview of Kerberos :



Kerberos operates by using a combination of client-server models, encryption and timestamps to authenticate users & services securely. The protocol involves three main components:

1. Client: The user or entity requesting authentication.
2. Server: The service or resource the client is trying to access
3. Key Distribution Center (KDC): A trusted entity that helps authenticate both clients and servers. The KDC is split into two parts:
 - the Authentication Service (AS) and
 - the Ticket Granting Service (TGS)

Kerberos Version 4 & Vs Kerberos Version 5:

Feature	Version 4	Version 5
Encryption Support	Limited (mostly DES)	Multiple algorithms (DES, AES, 3DES)
Ticket Format	Simple, limited functionality	Complex, extensible with renewable support
Cross-Realm Support	Limited	Improved, with better trust management
Name Format	Simple	Structured, more flexible
Replay protection	Basic	Enhanced
Delegation	Not supported	Supported
Ticket Renewal	Not supported	Supported
Security features	weaker	stronger, with enhanced protection.

Kerberos Version 5 is a significant improvement over version 4, providing enhanced security, flexibility & support for modern authentication requirements.

2. Explain in detail about DSA and DSS?

Ans: Digital Signature Algorithm : (DSA)

The digital signature algorithm is a federal information processing standard that governs digital signatures. The NIST suggested DSA in 1991 for use in the digital signature standard (DSS).

Global Public-Key Components

p prime number where $2^{L-1} < p < 2^L$ for $512 \leq L \leq 1024$ and L a multiple of 64; i.e., bit length of b/w 512 and 1024 bits in increments of 64 bits

q prime divisor of $(p-1)$, where $2^{159} < q < 2^{160}$; i.e., bit length of 160 bits

$$g = h^{(p-1)/q} \bmod p,$$

where h is any integer with $1 < h < (p-1)$ such that $h^{(p-1)/q} \bmod p > 1$

User's private key

x random or ~~super~~ pseudorandom integer with $0 < x < q$

User's public key

$$y = g^x \bmod p$$

User's Per-Message Secret Number

k = random or pseudorandom integer with $0 < k < q$

Signing

$$r = (g^k \bmod p) \bmod q$$

$$s = [k^{-1} (H(M) + xr)] \bmod q$$

$$\text{Signature} = (r, s)$$

Verifying

$$w = (g')^{-1} \text{ mod } q$$

$$u_1 = (H(M')w) \text{ mod } q$$

$$u_2 = (s')w \text{ mod } q$$

$$u = [(qu_1 y^{u_2}) \text{ mod } p] \text{ mod } q$$

$$\text{TEST : } v = x'$$

M = message to be signed

$H(M)$ = hash of M using SHA-1

M', x', s' = received versions of M, x, s

Digital Signature standard (DSS) :

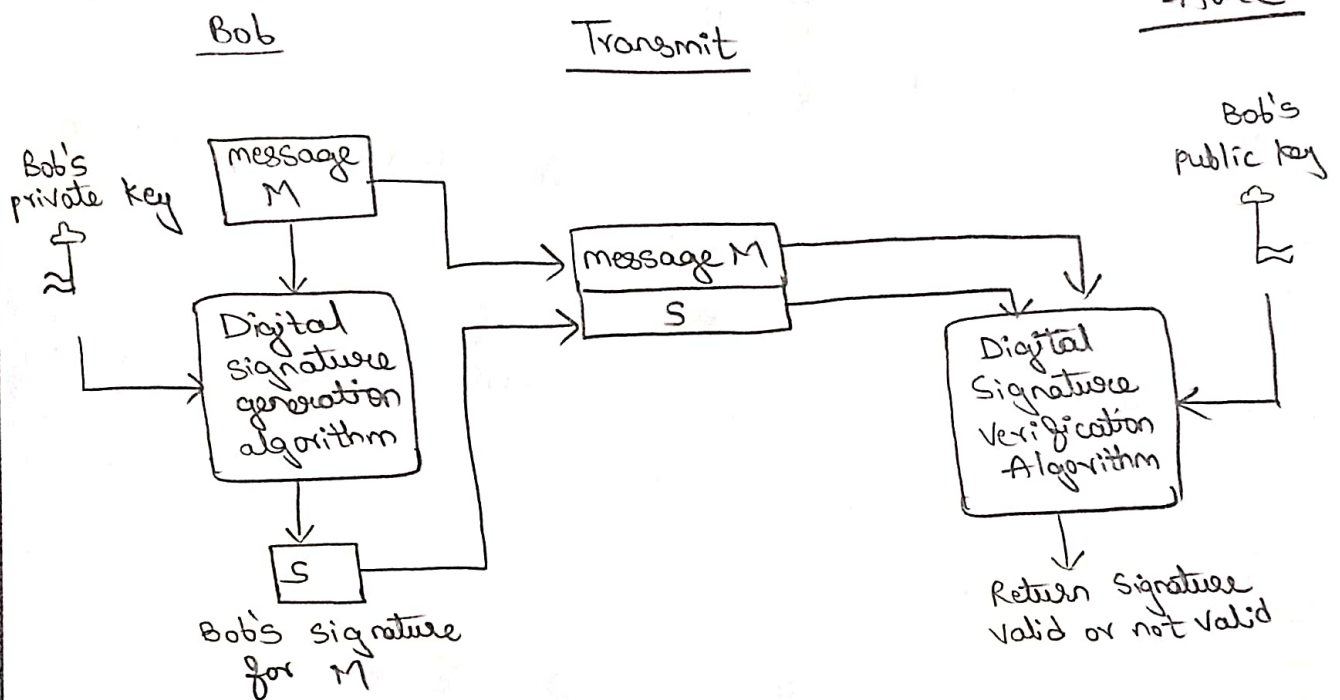


fig : Generic model of DSS

The DSS is a cryptographic standard established by the NIST in 1994 for generating & verifying digital signatures.

Working :

A hash of the message is computed. The sender's private key is used to generate the digital signature. The sign is sent along with the message. The receiver calculates the hash of the received msg. The sender's public key is used to decrypt & verify the sign, if matches it's become valid.